

Start Bioinfo RSG-ISCB

NEXTFLOW

Hands-on



Mateus Falco
2026

Qual sua experiência com bioinfo?

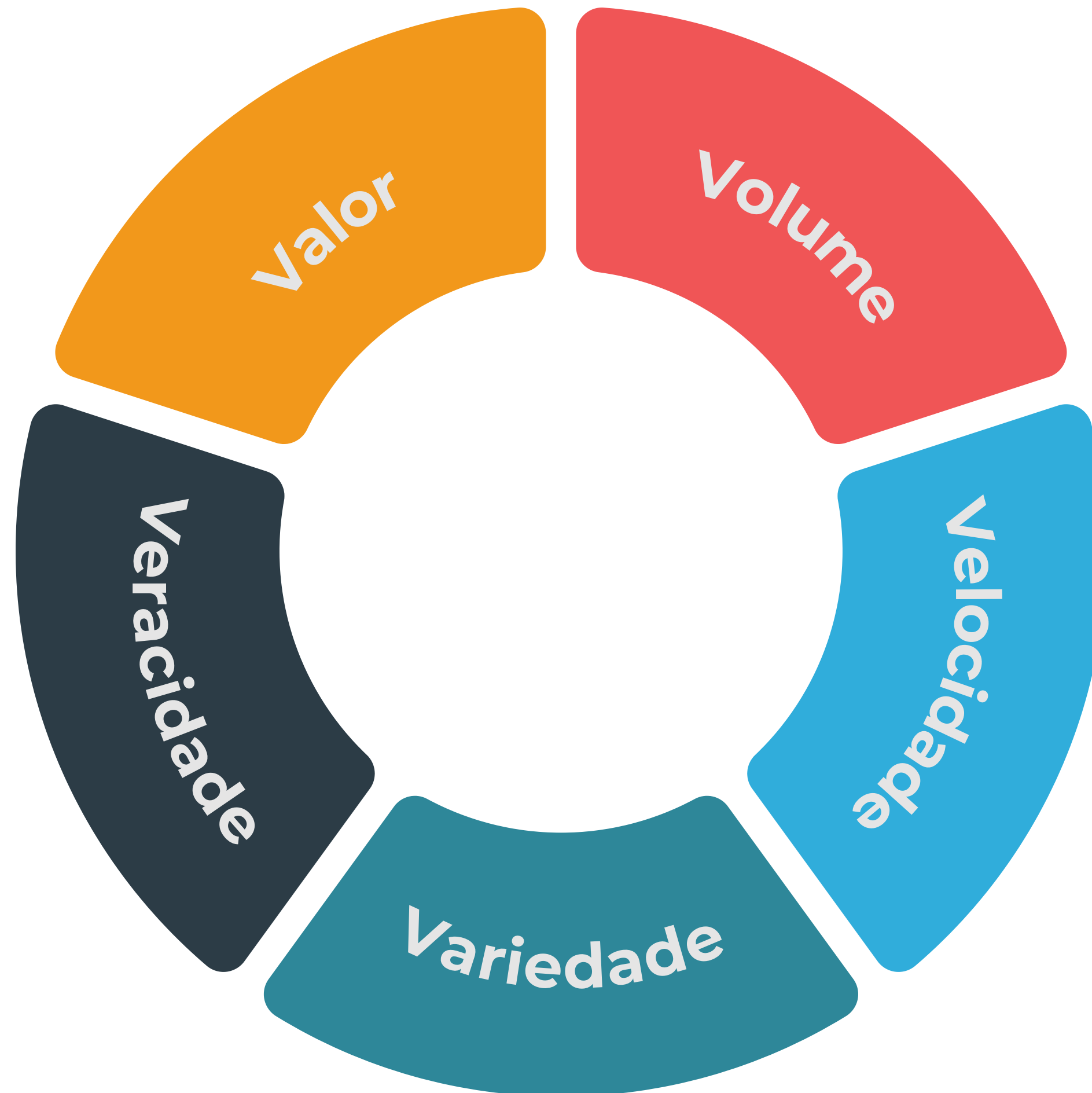


- Iniciante (6 meses)
- Intermediário (6 meses a 2 anos)
- Experiente (mais de 2 anos)



BEM-VINDOS A
ERA DO BIG DATA!

O que é o BIG DATA?



BIG DATA

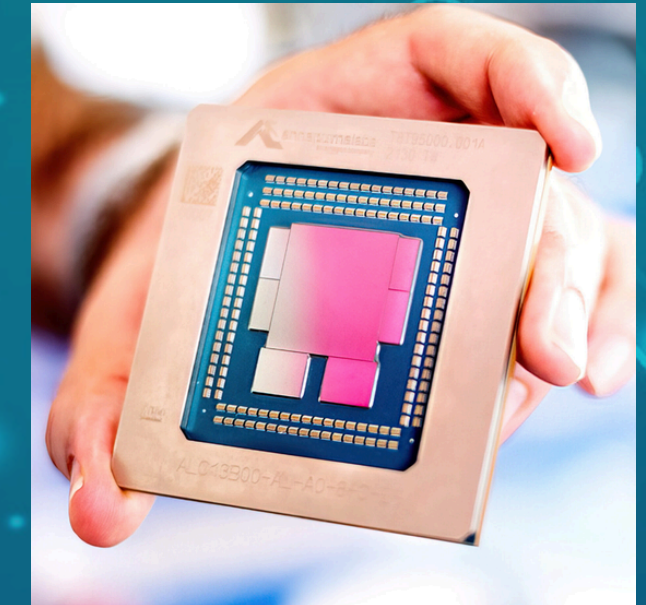
2024



The Institut Pasteur and AWS are analysing the world's DNA, using a public database

Institut Pasteur, a leading French virology research center, processed 20 petabytes of DNA data in record 30 hours, leveraging AWS Batch over a cluster of 2.18M AWS Graviton cores.

[Overview](#) | [Opportunity](#) | [Solution](#) | [AWS Services Used](#) | [Architecture Diagram](#)

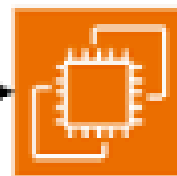


2.18 Million vCPUs Mobilised

The Institut Pasteur consequently turned to AWS to set up an appropriate infrastructure to meet the challenge of this massive processing. "Preparing the operations took almost a year, ultimately resulting in a calculation batch lasting just 30 hours," smiles Rayan Chikhi. But what a batch! During processing, AWS mobilised considerable resources, which reached 2.18 million vCPUs at peak for the [AWS Graviton](#) instances. As a comparison, we reckon that it would have taken a desktop computer nearly 30 million hours, or 3,400 years, to carry out such a calculation."



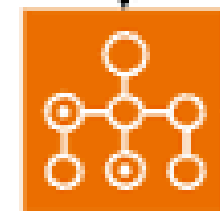
Operation



Bastion

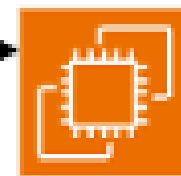
Inspection >

Start / Stop >

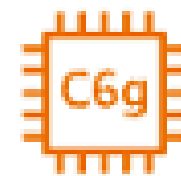


AWS Batch

Manage >



Amazon EC2
Spot Instances



AWS
Graviton

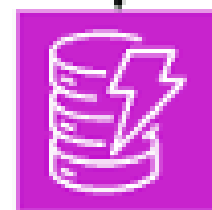
Container image

Business metrics flow

Technical metrics / logs flow



Amazon Elastic Container Registry



Amazon DynamoDB

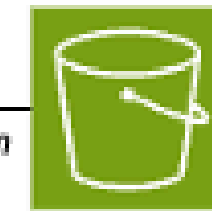


Amazon Managed Grafana

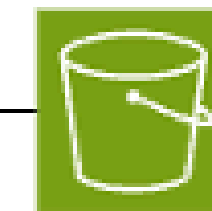


Amazon CloudWatch

< Load Data from

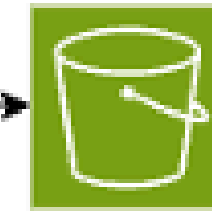


Amazon S3
Sequence Read Archive (SRA)
Open Data



Amazon S3
Mirror of NCBI RefSeq database

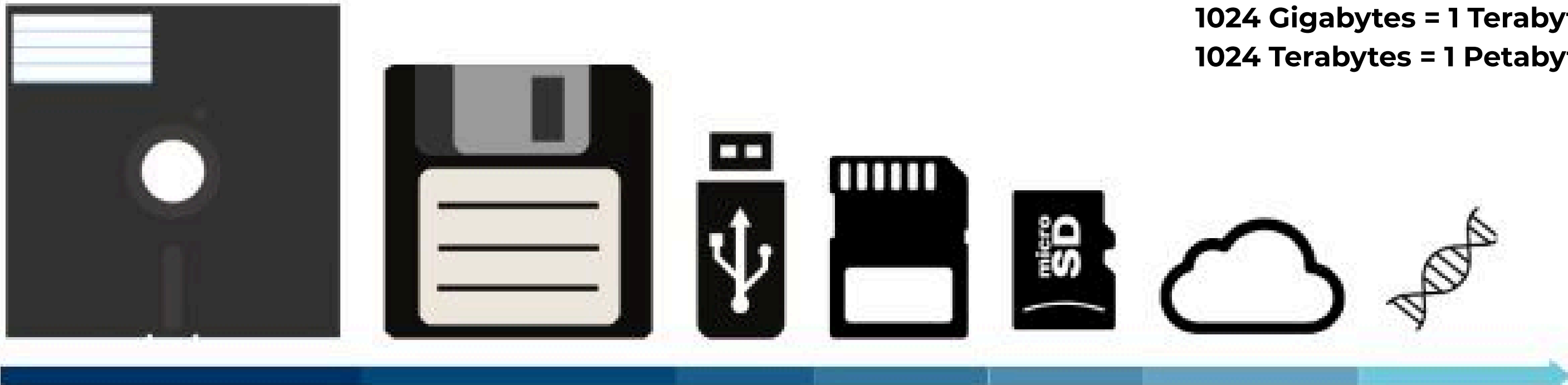
Write Data to >



Amazon S3
Logan / IndexThePlanet Open Data

Evolution of Data Storage

- 8 bits = 1 byte
- 4 bits = 1 nibble
- 1024 bytes = 1 kilobyte (KB)
- 1024 kilobytes = 1 Megabyte (MB)
- 1024 Megabytes = 1 Gigabyte (GB)
- 1024 Gigabytes = 1 Terabyte (TB)
- 1024 Terabytes = 1 Petabyte (PB)



	8 inch Floppy disk	5 1/2 inch Floppy disk	Pen Drive	SD Card	Micro SD Card	Cloud	DNA
Commercially Available	1971	1976	2000	2010	2010	2009	Now
Maximum Capacity	1.2 MB	1.2 MB	2 TB	256 GB	128 GB	~5 ZB	~ ∞
Cost per GB in Rupees	~ 40,000						~ 1 Re by 2023

DNA Storage

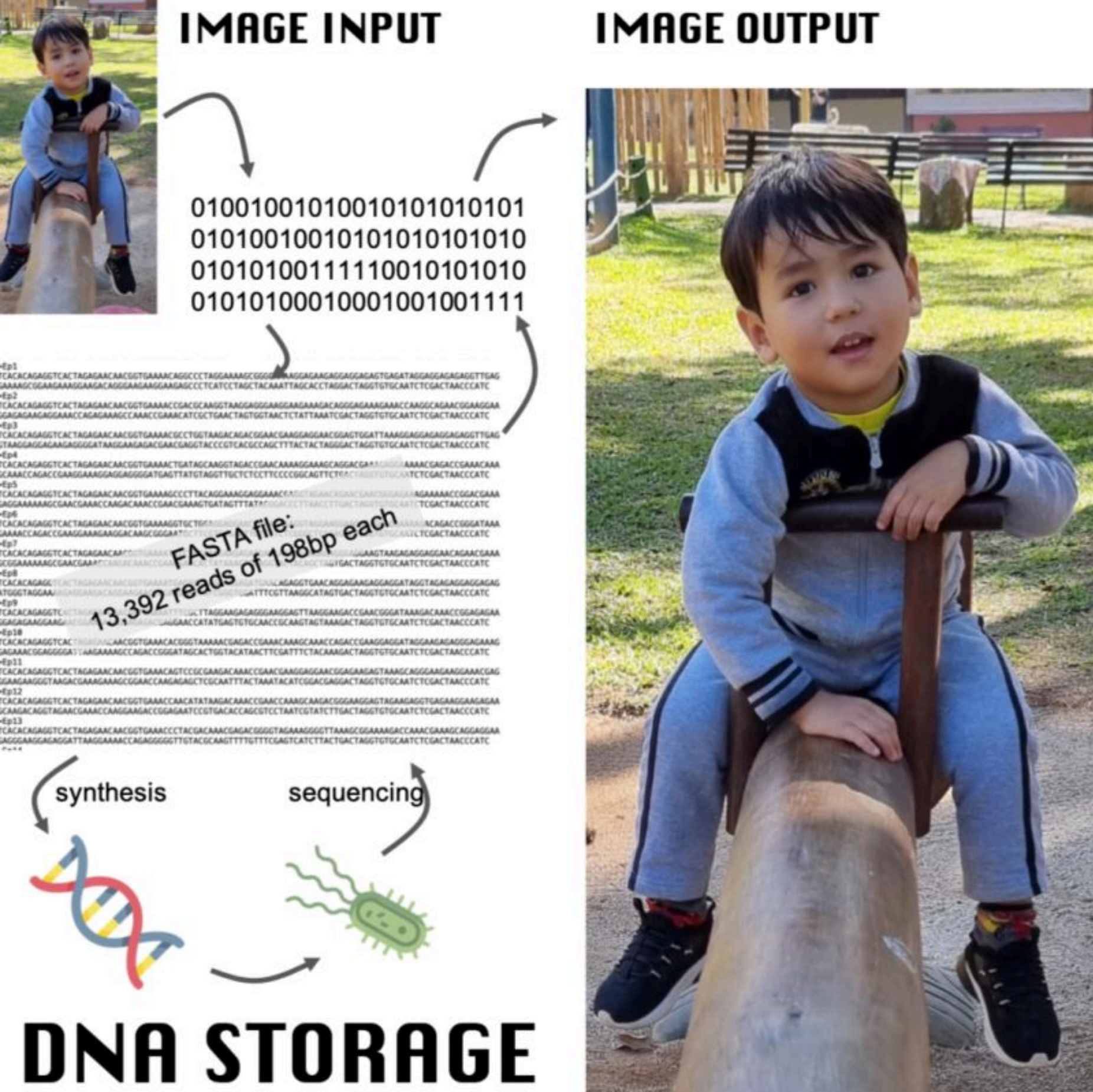
A (00)C (10)

T (01)G (11)

1g DNA ~ 215 PB



Prof. Dr. Helder Nakaya
USP/ Pesquisador Hospital IsraelitaAlbert Einstein



Qual o tamanho do genoma humano (diploide) sequenciado?

0 1,1 GB

0 3,3 GB

0 6,6 GB

“E como rodar essa quantidade de dados?”



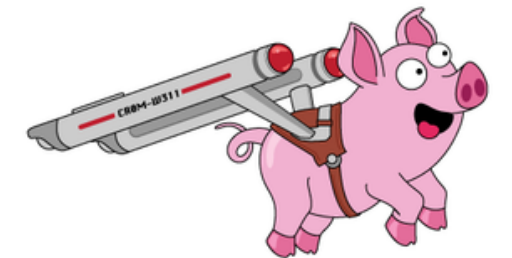
Sequencial

VS

 **nextflow**

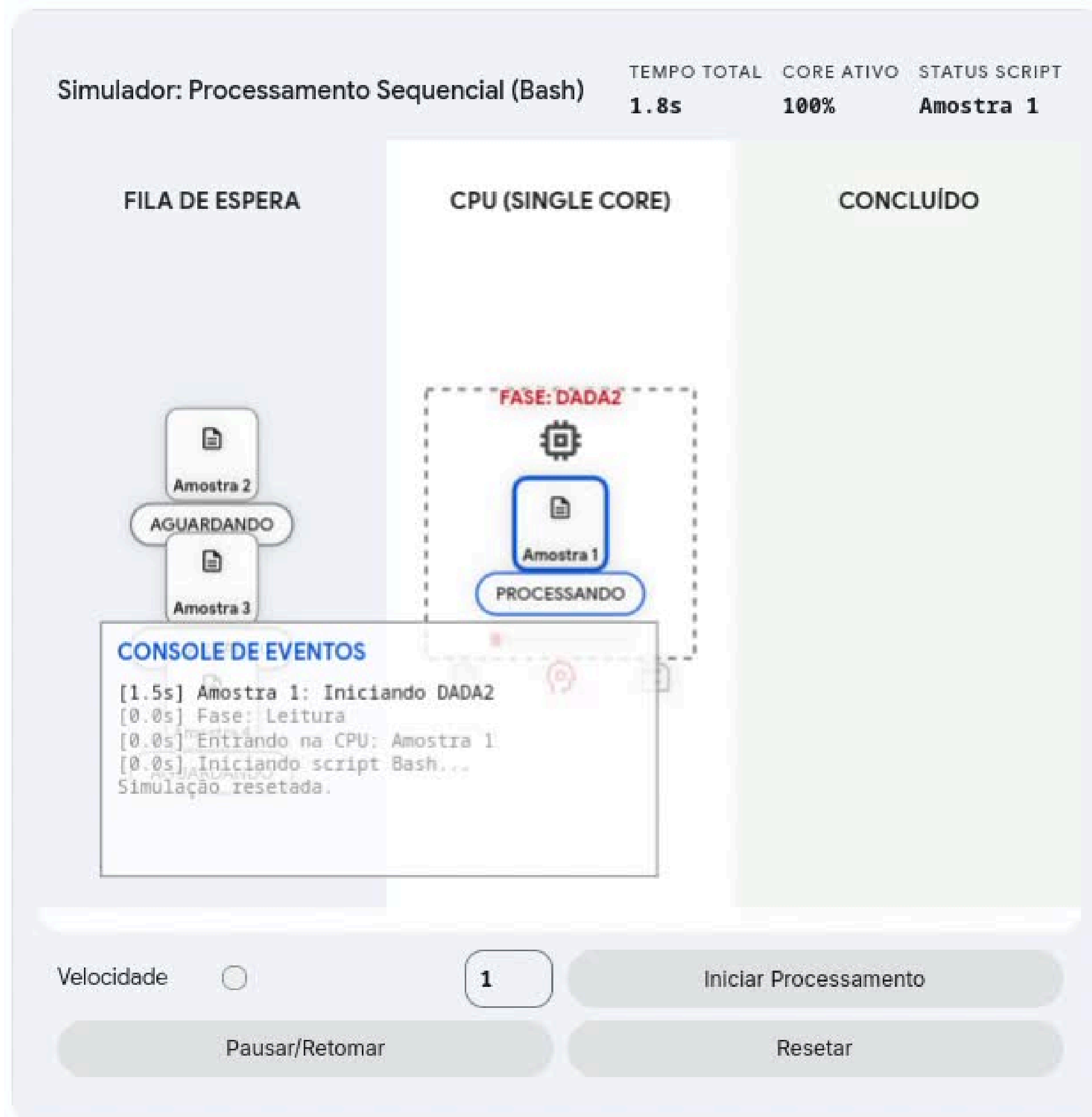


snakemake



Paralelo

Sequencial



Paralelo

DAG/Dataflow



Comparação

1 core / 4 cores

Pipeline Execution: Sequential vs Parallel

BASH RUNTIME
00:00.00

NEXTFLOW RUNTIME
00:00.00

SPEEDUP
0x

Bash Pipeline (Sequential / 1 Core)

Timer: 00:00.00

S00_Human	FastQC	Cutadapt	DADA2
S01_Mouse23	FastQC	Cutadapt	DADA2
S02_Mouse7	FastQC	Cutadapt	DADA2
S03_Mouse27	FastQC	Cutadapt	DADA2

Nextflow Pipeline (Parallel / 4 Cores)

Timer: 00:00.00

S00_Human	FastQC	Cutadapt	DADA2
S01_Mouse23	FastQC	Cutadapt	DADA2
S02_Mouse7	FastQC	Cutadapt	DADA2
S03_Mouse27	FastQC	Cutadapt	DADA2

Start Sequential (Bash)

Start Parallel (Nextflow)

Reset Simulation

	 snake make	 nextflow	 {wdl}	 TOIL	 COMMON WORKFLOW LANGUAGE	 robin	 COMMON WORKFLOW LANGUAGE	Bpipe	 Galaxy PROJECT	 argo	 Apache Airflow
Readable/Popular Syntax	✓	✗	✗		✗		✗	⚠	✓	✗	⚠
Lightweight & Fast to Prototype	✓	⚠	✗		✗		✗	✓	⚠	⚠	✗
Reproducibility (envs / checksums)	✓	✓	✓		✓		✓	⚠	✓	✓	⚠
Container Support (Docker/Singularity)	✓	✓	✓		✓		✓	⚠	✓	✓	✓
Cloud-Native Execution Support	⚠	✓	✓		✓		⚠	✗	✗	✓	✓
HPC / Cluster Integration	✓	✓	✓		✓		✓	✓	⚠	✓	⚠
Robust Run Resumption	⚠	✓	✓		✓		✓	✗	✓	✓	✓
Modular Workflow Structure	⚠	✓	✓		✓		✓	⚠	✗	✓	✓
Auditable / Structured Workflows	⚠	⚠	✓		✓		✓	✗	✓	✓	✓
Support for Regulated Environments	✗	⚠	✓		✓		✓	✗	✓	⚠	✗
Strong Community Support	✓	✓	⚠		✓		⚠	⚠	✓	✓	✓
Easy to Debug & Maintain	✓	⚠	✗		✗		✗	✓	✓	✗	⚠
Standard Compliance / Interoperability	⚠	⚠	⚠		✓		✓	✗	✓	⚠	⚠
Declarative Language	⚠	✓	✓		✓		✓	✗	✗	✓	✓
Best Suited for Exploratory Research	✓	⚠	✗		✗		✗	⚠	✓	✗	✗
Well-Suited for Education / Training	⚠	✗	✗		✗		✗	⚠	✓	✗	✗
Good for Kubernetes-Native Environments	✗	✓	✗		✗		✗	✗	✗	✓	✓
Low Memory / Runtime Overhead	✓	✗	⚠		✗		⚠	✓	⚠	⚠	⚠
Git Integrated	✓	✓	✗		✗		✗	⚠	✗	✗	⚠
Mature Ecosystem for Interoperability	✗	✗	⚠		✓		✓	⚠	✗	✗	✗
Platform-Agnostic Portability	✗	⚠	⚠		✓		✓	✓	✓	✓	⚠

✓ = strong support ⚠ = partial or situational ✗ = not the strongest of strengths

Cristina Guijarro-Clarke (Linkedin)

「\ (ツ) /」

IT WORKS

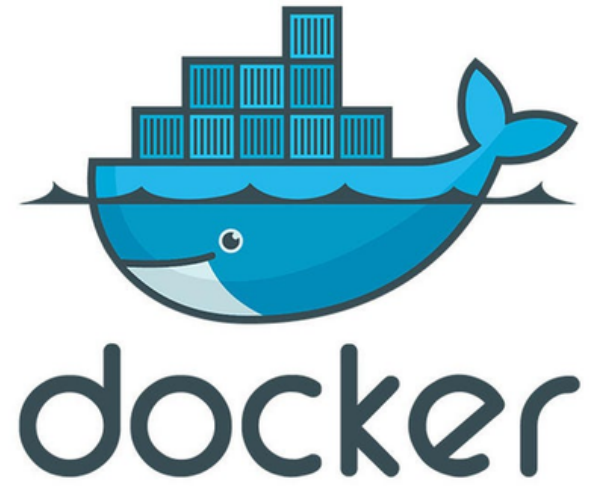
on my machine

Alguém já passou por isso?

0 SIM

0 NÃO

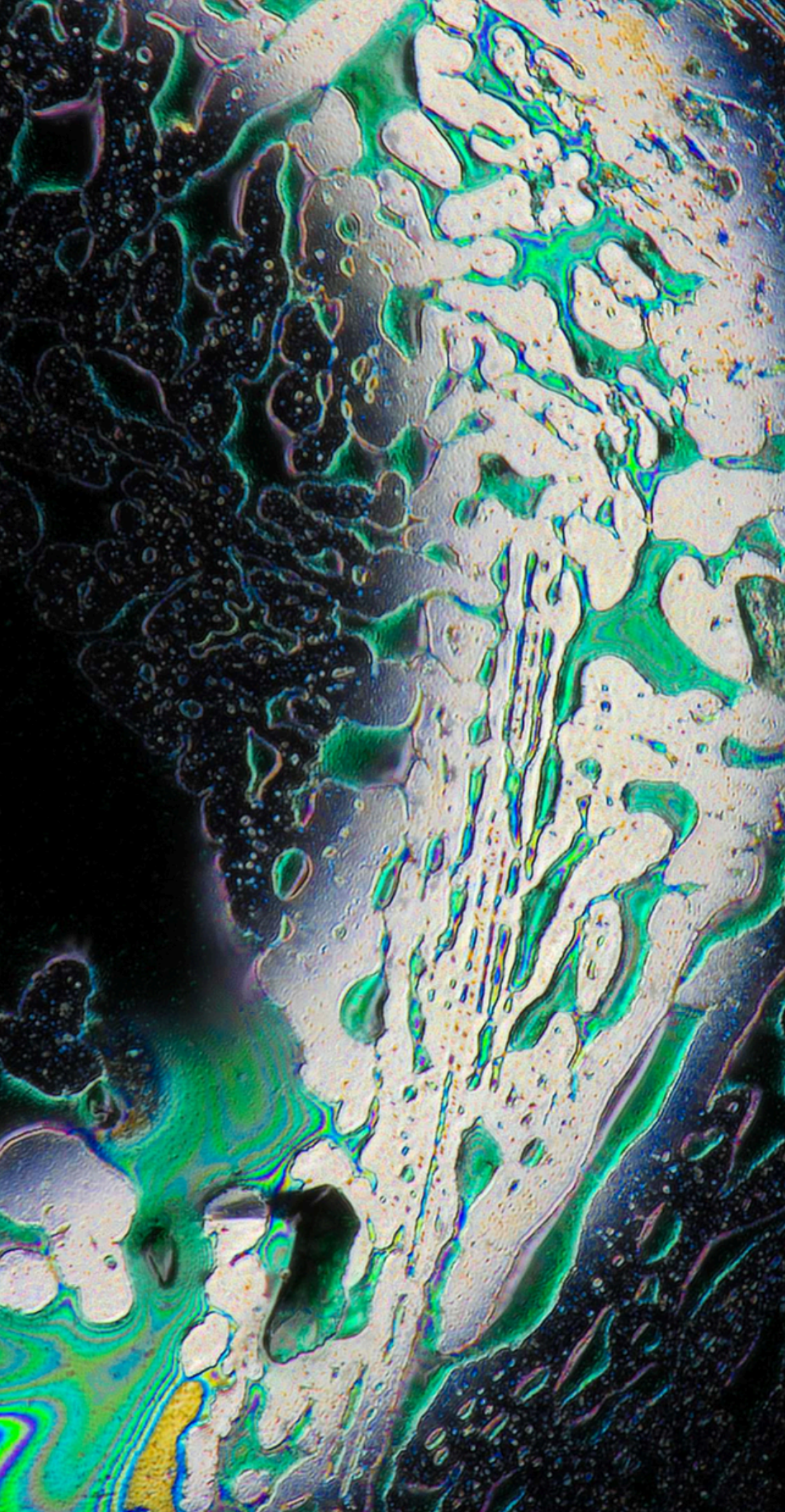
Containerização



Comparação

VM vs Container



A vertical strip on the left side of the image featuring a marbled paper pattern with swirling green, blue, and white colors.

Nextflow | Dataflow



Institucional

Fundadores, apoiadores e números



Nextflow

25,000+
Estimated Nextflow users

9,000
Seqera Cloud users

1,900
Nextflow community
open-source contributors

150+
Seqera Enterprise
customers

100
High-quality, curated nf-
core pipelines

160,000
Nextflow monthly
downloads

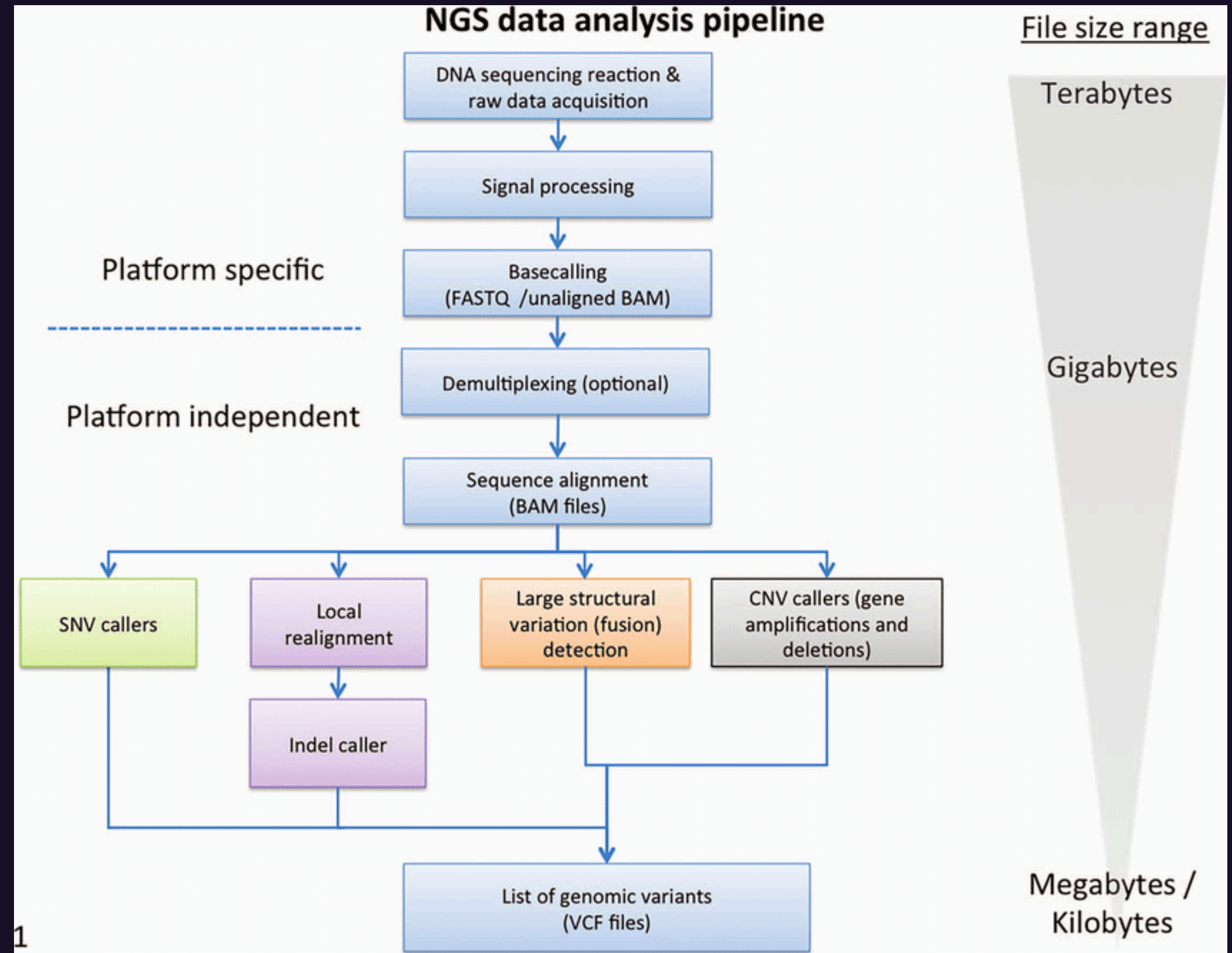
85
Seqera employees

7,500+
Citations for Nextflow, nf-
core, & MultiQC



NGS Workflow

From Terabytes to Megabytes



Workflows

Using computers to collect, store, analyze, and disseminate data and information



Large files

> 100 GB for one raw human genome...



Many languages

Bash, Python, R, PERL...



Complex interactions

Networks of software and their dependencies...



Reprodutibilidade

Problemas de reprodutibilidade são como *icebergs*



Primeiro, tentamos executar novamente a análise com o código e os dados fornecidos pelos autores.

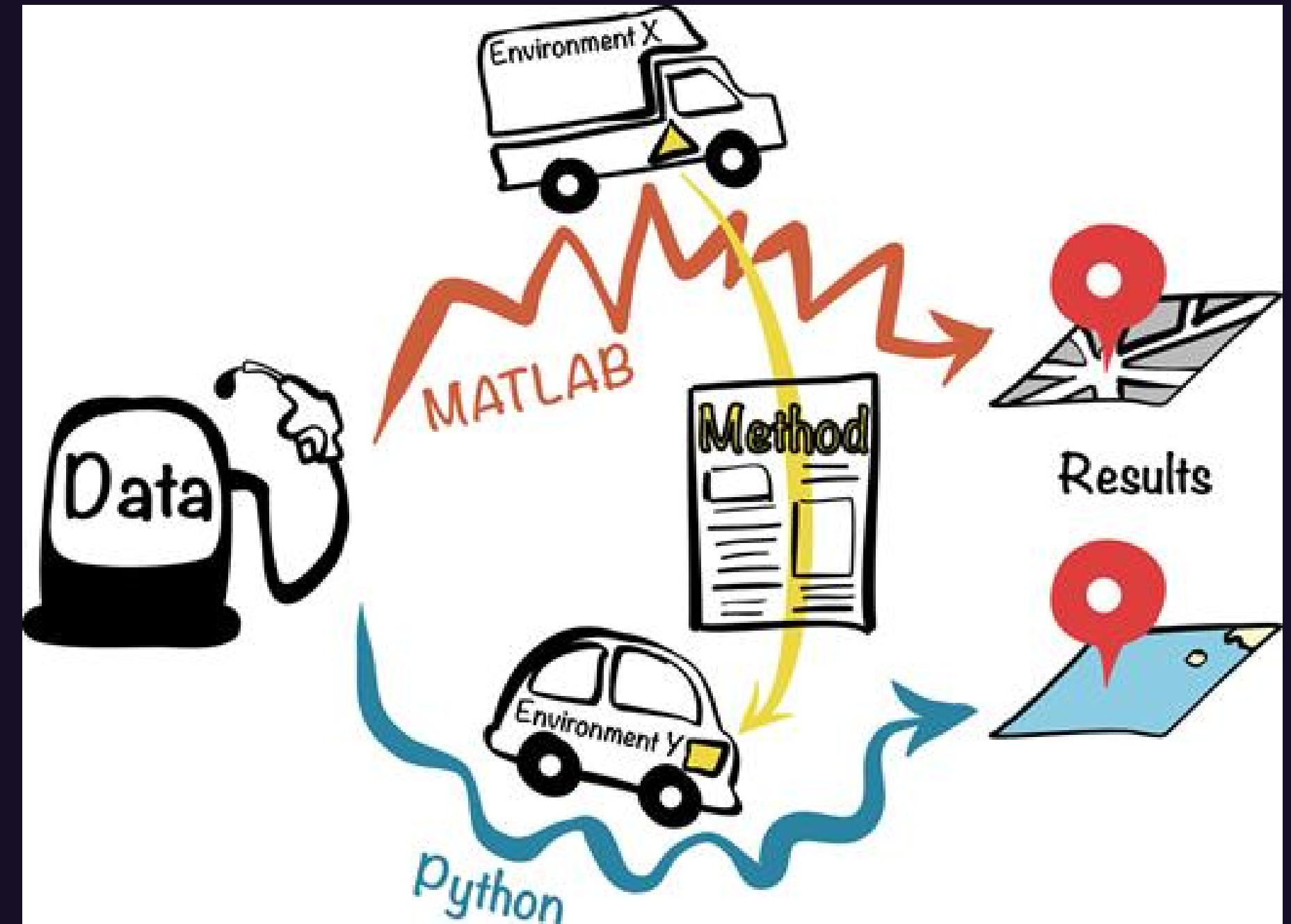
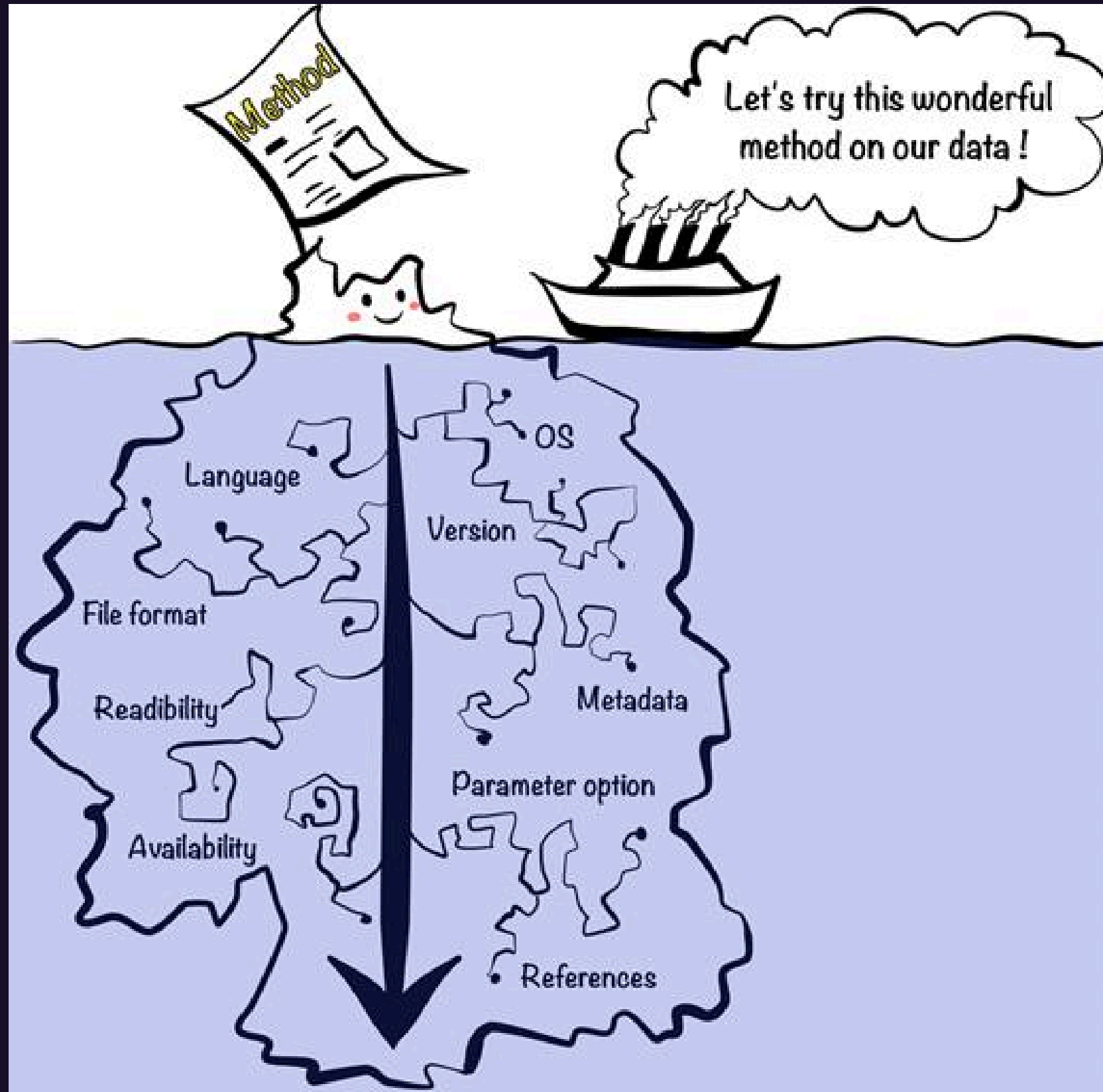
Em segundo, reimplementamos todo o método em um pacote Python...



[Experimenting with reproducibility: a case study of robustness in bioinformatics Kim et al., GigaScience \(2018\).](#)

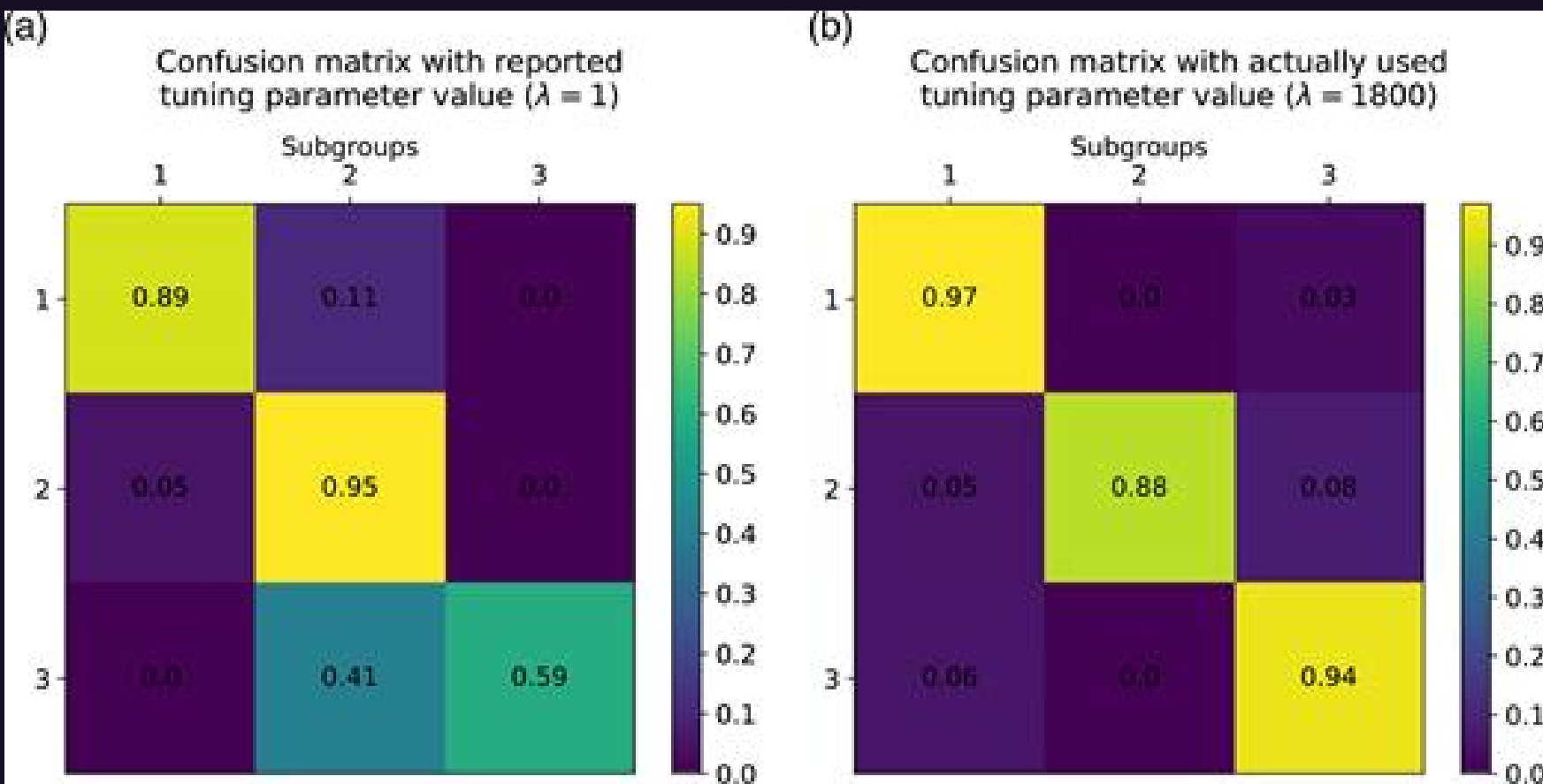
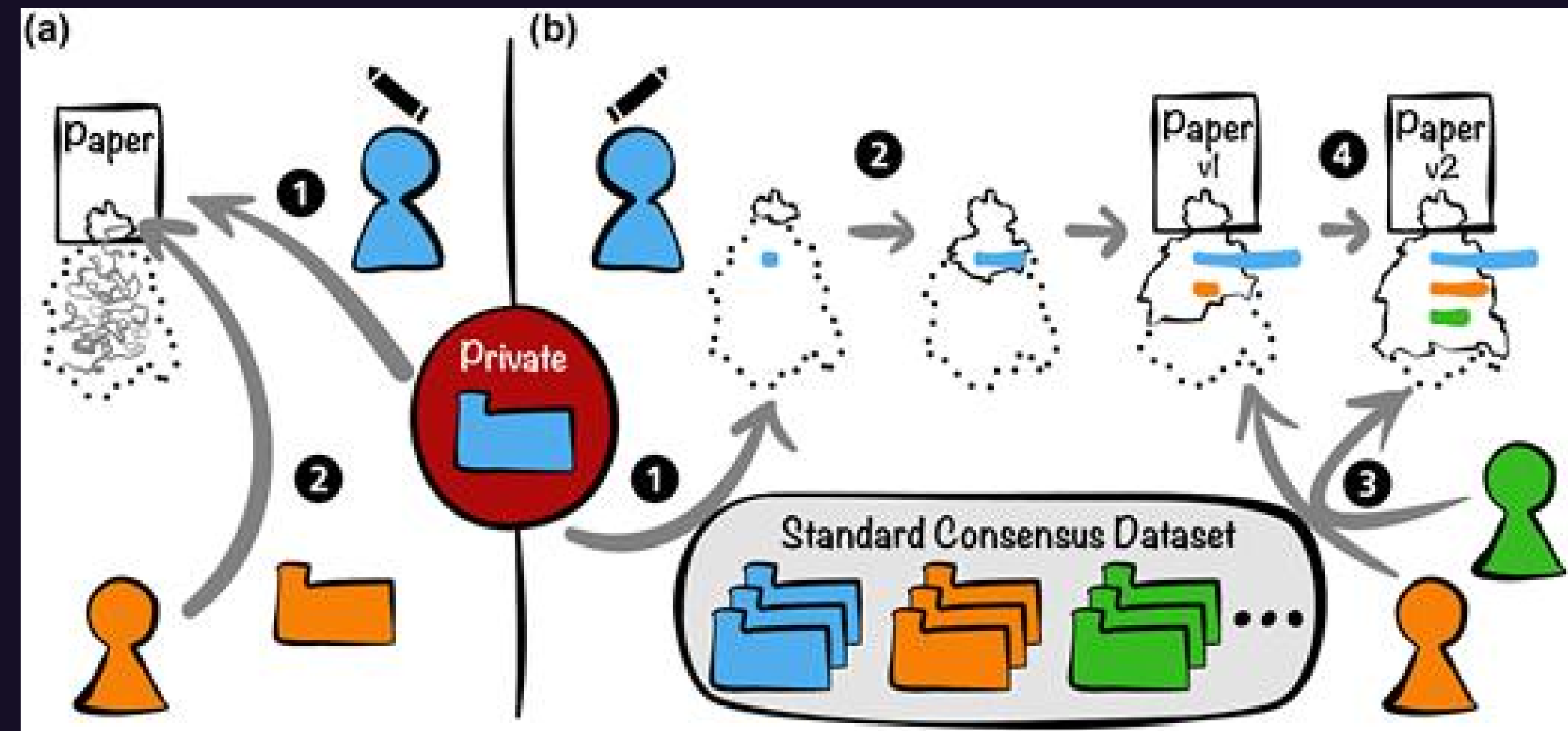


Reprodutibilidade



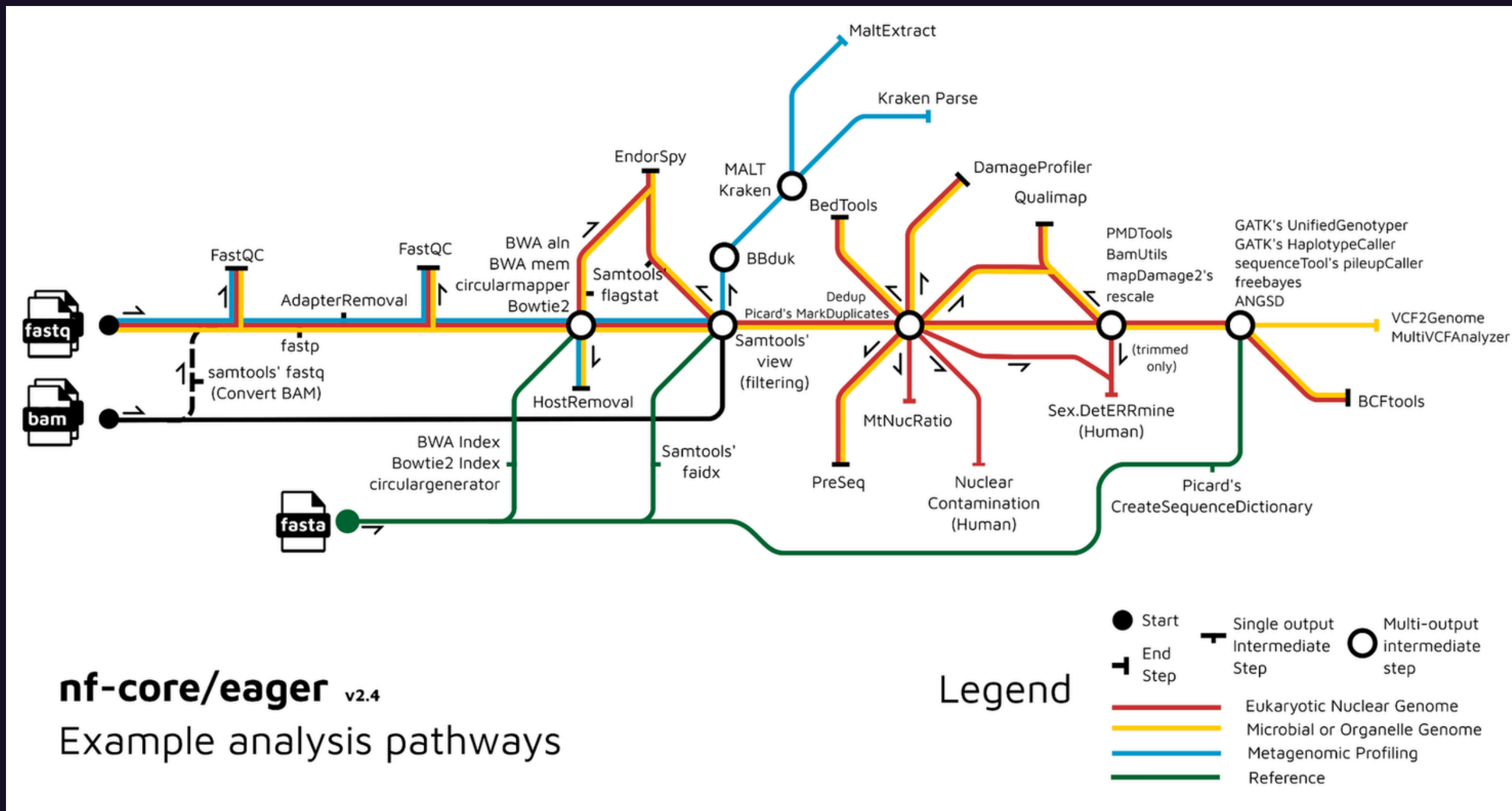
Reprodutibilidade

Solução



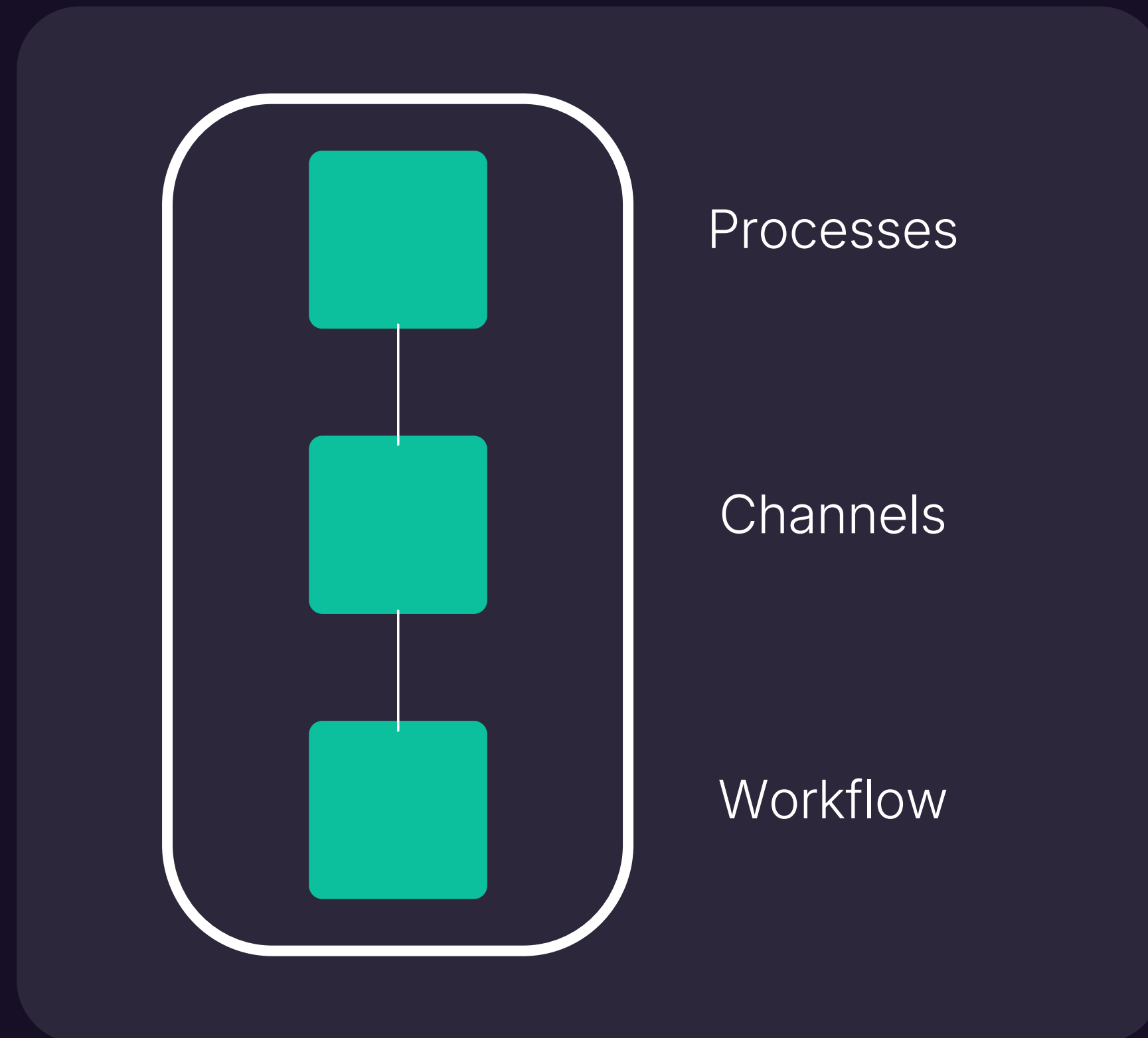
Resultado

Managing modern workflows is complicated



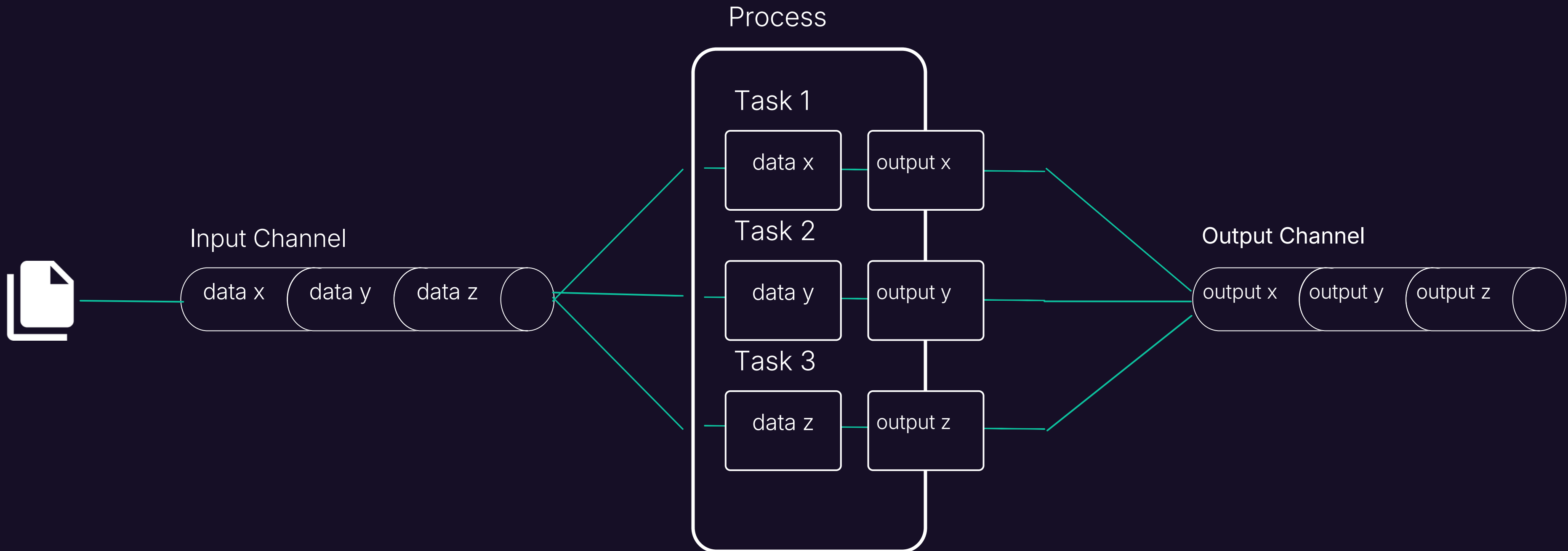
Nextflow

A reactive workflow framework and a programming DSL2



Nextflow

A reactive workflow framework and a programming DSL2



Nextflow

A reactive workflow framework and a programming DSL

```
input:
  path input

output:
  path "*_fastqc.{zip,html}"

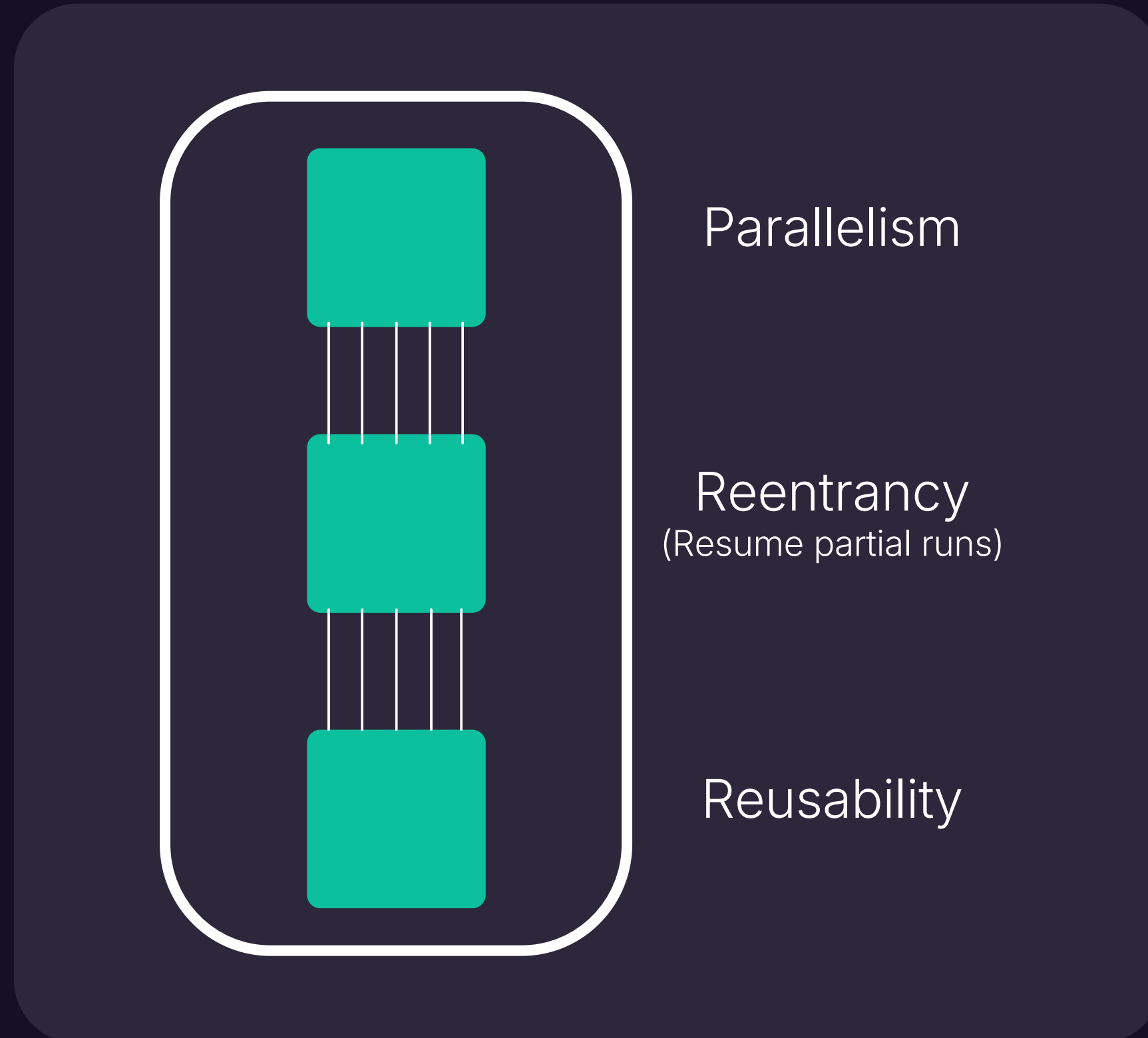
script:
  """
  fastqc -q $input
  """
}

workflow {
  script:
    """
```



Nextflow

A reactive workflow framework and a programming DSL



Nextflow

Nextflow is a language, a runtime, and a community

 **nextflow**
pipeline

Write code
in any language



Orchestrate tasks with
dataflow programming



Define software dependencies
via containers



Built-in version control with
Git



 **nextflow**
runtime

Task orchestration
and execution



Nextflow

Nextflow is a language, a runtime, and a community



Reproducible

Integration with code management tools, with versioned releases.



Portable

Docker, Singularity, Conda, works with most compute environments.



Scalable

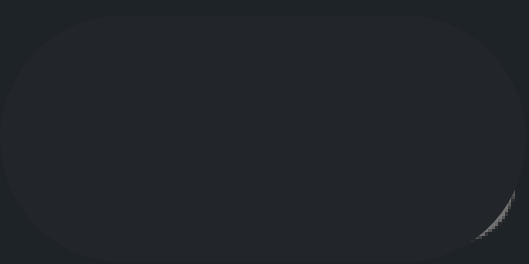
5 samples on your laptop, 5k on an HPC or 5 million in the cloud.



nf-core 

nf-core

A community effort to collect a curated set
of analysis pipelines built using Nextflow.



nf-core

A community effort to collect a curated set of analysis pipelines built using Nextflow

8k+

Slack
users

2k+

GitHub
contributors

120+

GitHub
repositories

40k

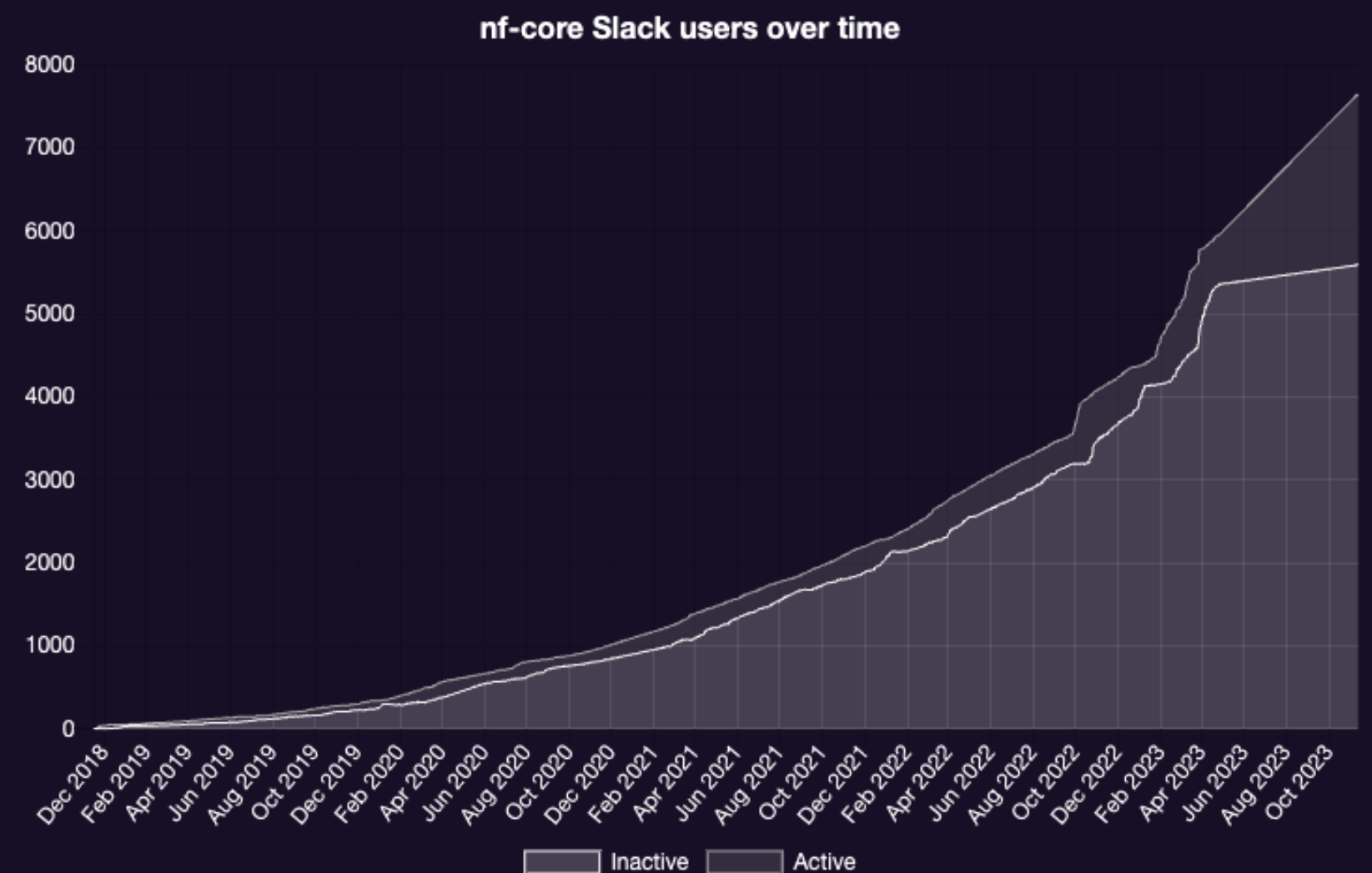
GitHub
commits

16k+

Pull
requests

7k+

GitHub
issues



nf-core principles

Principles that guide the nf-core community



Cooperation

Develop with the community



Standards

Use a common template



Collaboration

No duplicate pipelines within nf-core



Helper Tools

Tools built for everyone



Compatibility

Tools work for any Nextflow pipeline



Components

Collaborate on component-level



nf-core components

Pick and choose which component you need



Pipelines

>100 pipelines and a base template



Subworkflows

>55 subworkflows



Modules

>1150 modules



Linting

Choose conventions to test for consistency



Schema

Validation, channels and user interface



Tooling

Development and deployment



nf-core/tools

Command line tools to help you build your pipeline with ease



Pipelines

Create from template,
sync to get updates



Subworkflows

Create, install and
update



Modules

Create, install, update,
patch, test



Schema

Build your pipeline
schema with a GUI



Linting

Test nf-core standards
and best practices



Download

Fetch with singularity
images for offline use



nf-core/pipeline

Running pipeline

```
alexanderpeltzer@Alexanders-MBP ~/I/n/bacass> nextflow run . -profile test,docker --dump-channels
N E X T F L O W ~ version 19.04.1
Launching `./main.nf` [marvelous_booth] - revision: 3a01436899
[2m-----

      ,--. ,-.
     /,-.-.-~'
    }  {
   \-.-,-\-,
    .-.-.

nf-core/bacass v1.0dev
-----
Pipeline Name      : nf-core/bacass
Run Name           : marvelous_booth
Skip Kraken2       : true
Kraken2 DB         :
Extra Unicycler arguments: --no_correct --no_pilon --kmers 33
Extra Prokka arguments: --fast
Max Resources      : 6 GB memory, 2 cpus, 2d time per job
Container          : docker - nfcore/bacass:dev
Launch dir         : /Users/alexanderpeltzer/IDEA/nf-core/bacass
Output dir         : ./results
Working dir        : /Users/alexanderpeltzer/IDEA/nf-core/bacass/work
Script dir         : /Users/alexanderpeltzer/IDEA/nf-core/bacass
User               : alexanderpeltzer
Config Profile     : test,docker
Config Description: Minimal test dataset to check pipeline function
[2m-----
[warm up] executor > local
executor > local (4)
[ab/b76142] process > get_software_versions [100%] 1 of 1 ✓
[af/f0ecba] process > output_documentation [100%] 1 of 1 ✓
[0c/29da89] process > trim_and_combine [ 0%] 0 of 2
Staging foreign file: https://github.com/nf-core/test-datasets/raw/bacass/ERR044595_1M_2.fastq.gz
Staging foreign file: https://github.com/nf-core/test-datasets/raw/bacass/ERR064912_1M_2.fastq.gz
█
```



Runnig Nextflow/nf-core

Organisation running nf-core pipelines around the world



nf-NEXUS: A scalable, cloud-native Nextflow framework for strict reproducibility analytical pipelines

```
N E X T F L O W ~ version 25.10.4

Launching `NEXUS.nf` [silly_gauss] DSL2 - revision: 02e546b166

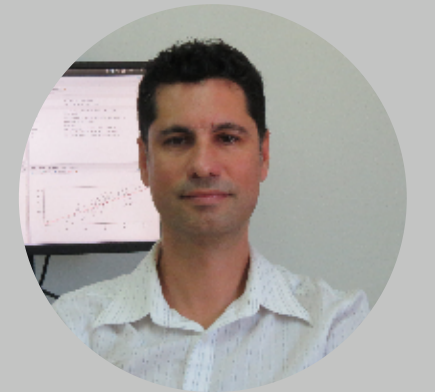
=====
nf-NEXUS: ████████ analytical workflows replication pipeline
=====
```



Marcel Ribeiro-Dantas. Nextflow and nf-
core Developer Advocate for Latin America
at Seqera Labs



João Cavalcante
Bioinformatic Developer



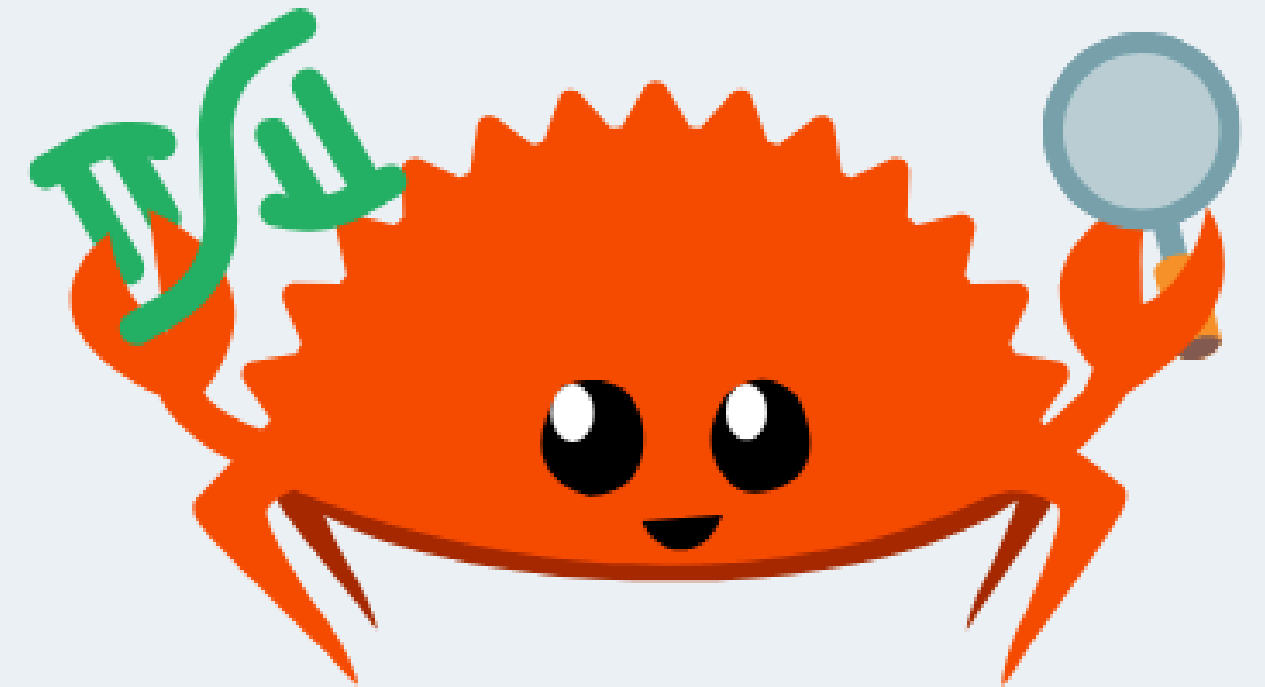
Prof. Rodrigo Dalmolin (UFRN)

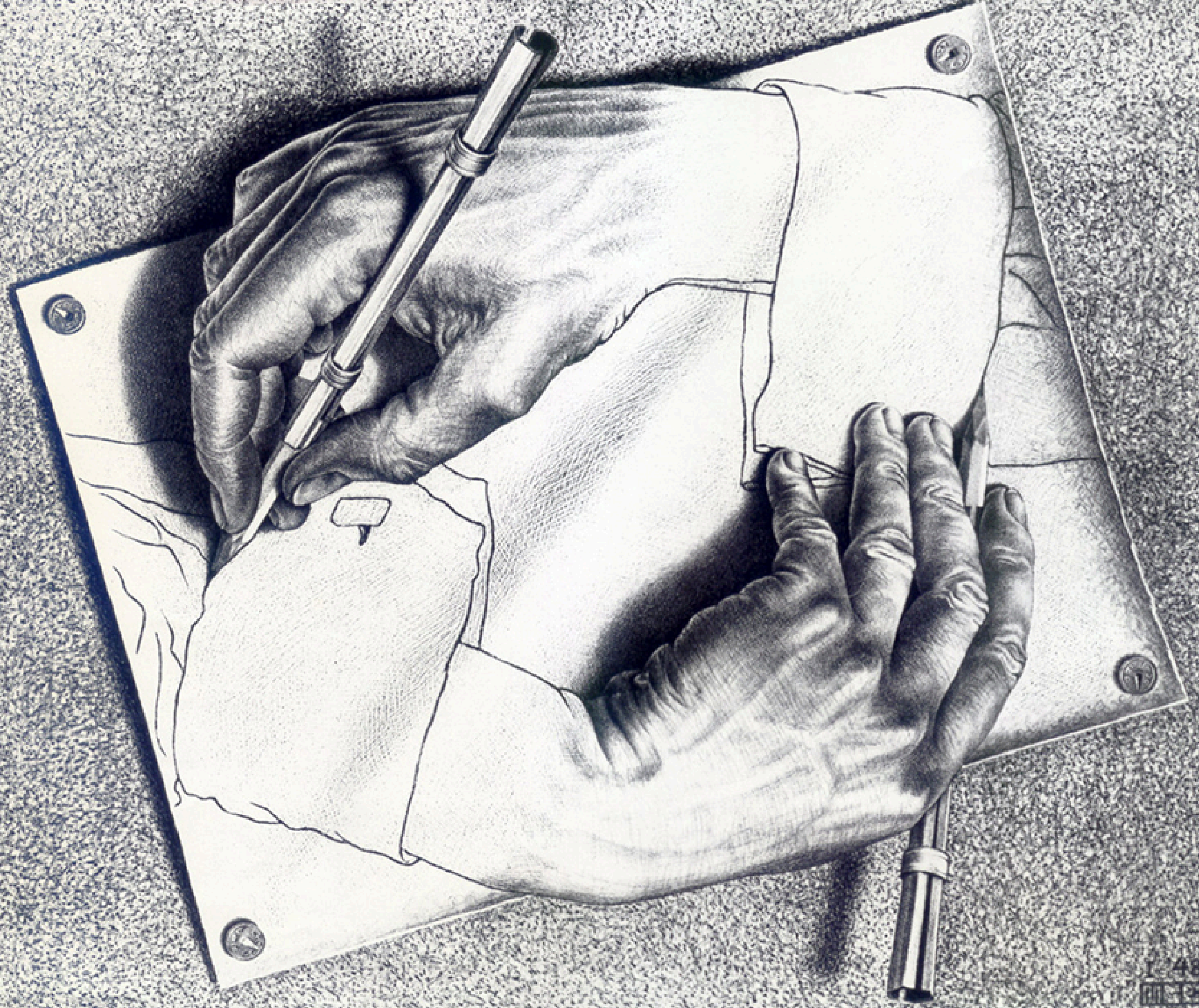
Welcome to RustQC

A fast reimplementation of [dupRadar](#), [featureCounts](#), [RSeQC](#), [Qualimap](#), [preseq](#), and [samtools](#) in Rust, delivering matching results much faster.



Run time for a large paired-end RNA-seq BAM (~186M reads) on AWS. RSeQC TIN alone takes 9h 45m; RustQC completes everything in 14m 54s.





Hands-on

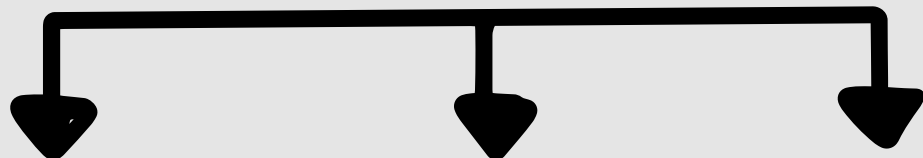
M.C. Escher, *Drawing Hands*, 1948



19022



Dia 0



Dia 7



Dia 23

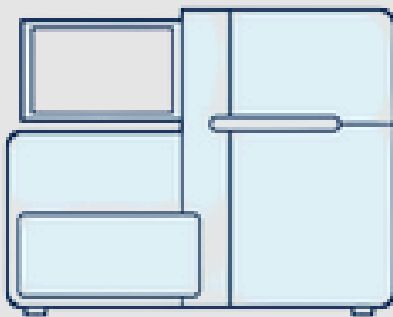


Dia 27

Camundongo #2

FMT

NGS

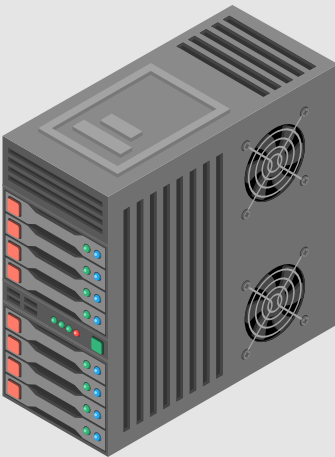


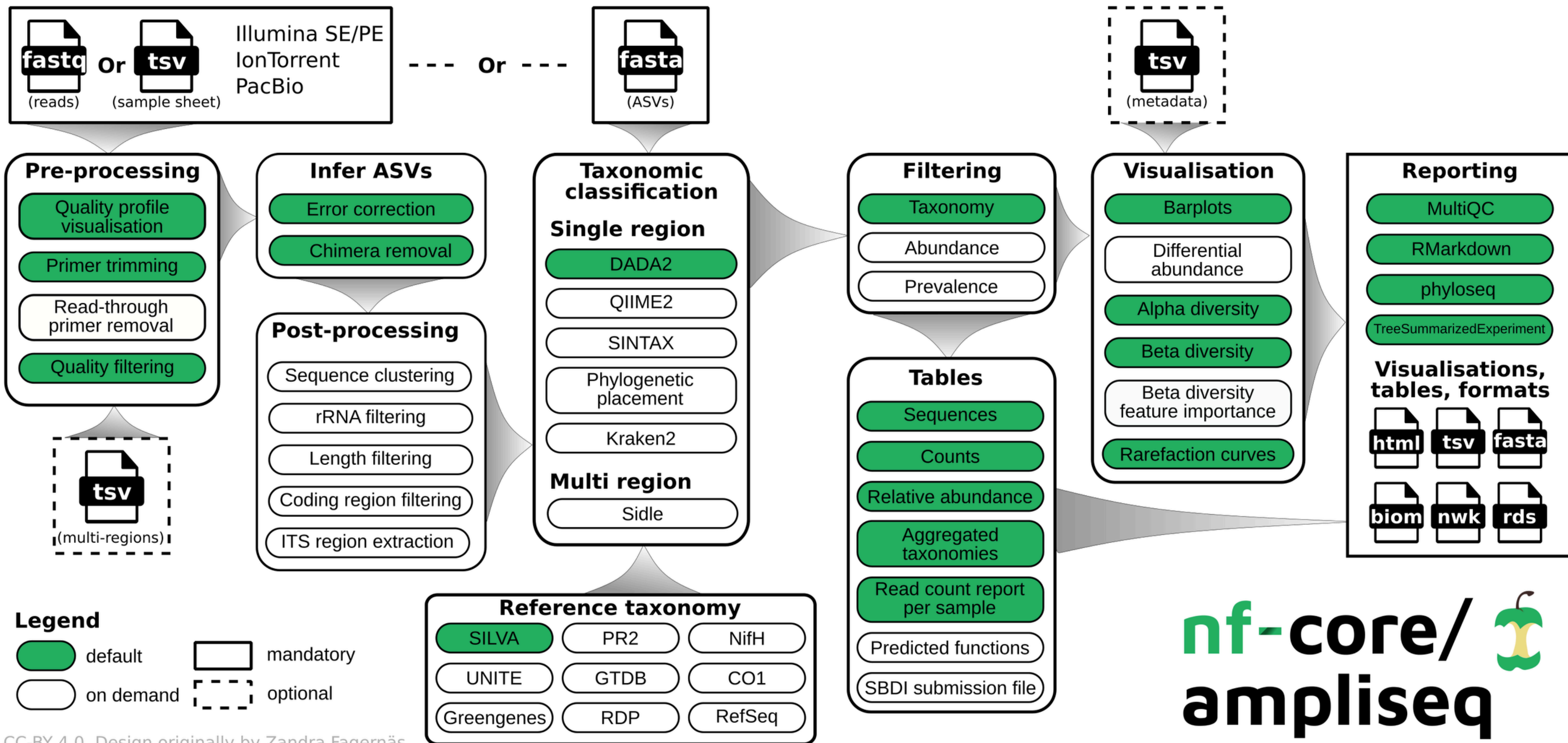
Data

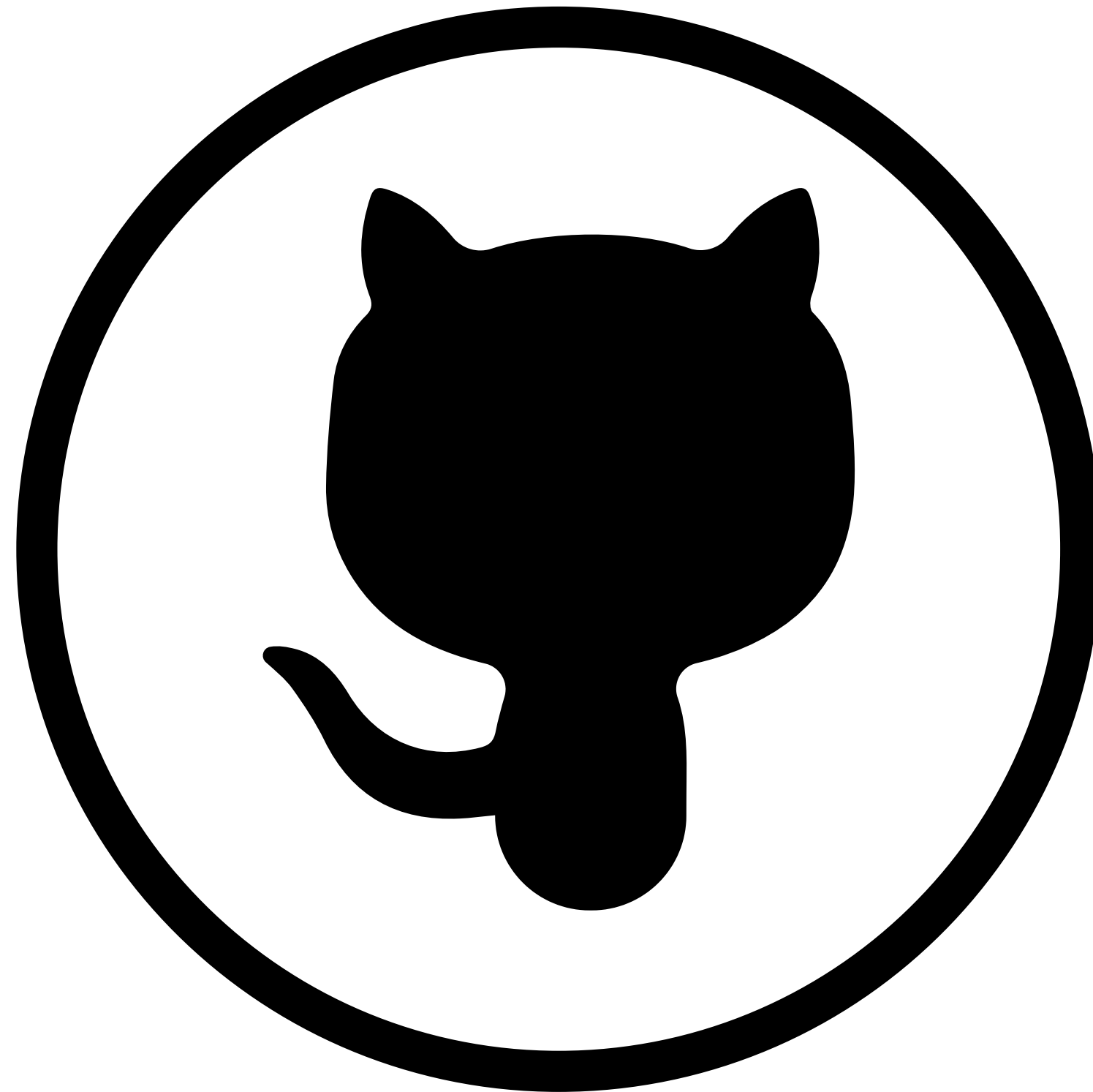
Library1_2_S1_L001_R1_001.fastq.gz
Library1_2_S1_L001_R2_001.fastq.gz
Library1_2_S1_L001_I1_001.fastq.gz
Library1_2_S1_L001_I2_001.fastq.gz
Library1_2_S1_L001_R1_001.fastq.gz
Library1_2_S1_L001_R2_001.fastq.gz
Library1_2_S1_L001_I1_001.fastq.gz
Library1_2_S1_L001_I2_001.fastq.gz



Server







Bora praticar!